

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 June 2025
Team ID	LTVIP2025TMID58466
Project Name	Resolvenow: Your Platform For Online Complaints
Maximum Marks	4 Marks

Technical Architecture:

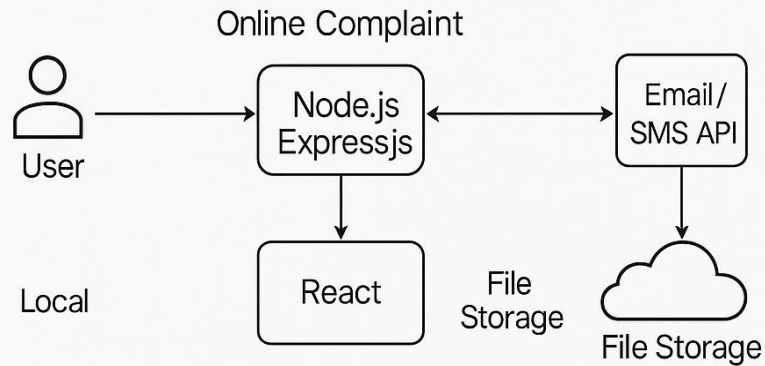
The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

Example Scenario – ResolveNow :-

John, a registered user, logs into the **ResolveNow** platform via Gmail. He notices a leaking pipe issue in his apartment and submits a complaint using the web interface, attaching a photo of the leak. The system automatically assigns his complaint to **Agent Sarah** based on the complaint category (plumbing). John receives an email confirmation with a tracking ID.

Later, John uses the **real-time chat feature** to follow up with Sarah, who updates the status as "In Progress." Once the issue is resolved, Sarah marks it as "Resolved." John receives an email notification and submits positive feedback through the dashboard.

ResolveNow: Online Complaint Management



Guidelines:

- Include all core processes as technology blocks (e.g., login, complaint, chat).
- Separate components into Local (frontend/backend) and Cloud (APIs, DB).
- Highlight all third-party interfaces like Email/SMS APIs, OAuth, etc.
- Specify data storage (e.g., MongoDB, file storage) and backup components.

Table-1 : Components & Technologies:-

S.No	Component	Description	Technology
1	User Interface	Web UI for users (React SPA), responsive layout	React.js, Bootstrap, Material UI, HTML, CSS
2	Application Logic-1	Backend logic for user registration, login, complaint handling	Node.js, Express.js
3	Application Logic-2	Real-time chat between user and agent	Socket.io, WebRTC
4	Application Logic-3	Complaint routing based on department or issue type	Custom logic (Node.js-based microservice)
5	Database	Store users, complaints, messages, feedback	MongoDB (NoSQL)
6	Cloud Database	Cloud-hosted DB with scalable storage and backup	MongoDB Atlas

7	File Storage	To store complaint attachments (images, docs)	Local storage or AWS S3 (optional upgrade)
8	External API-1	Email service for OTP/alerts	Nodemailer / SendGrid
9	External API-2	SMS service for complaint status alerts	Twilio API / Fast2SMS
10	Machine Learning Model	(Future Scope) Classify complaint urgency level	(To be added: sentiment analysis, NLP model)
11	Infrastructure	Hosting environment	Local server (Dev), Render / Heroku (Cloud)

Table-2: Application Characteristics:-

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Full-stack app built on open-source technologies	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	Authentication & secure access controls	JWT, HTTPS, bcrypt.js, Helmet.js, CORS
3	Scalable Architecture	Easily extendable microservice-based model with REST API	3-tier architecture (UI → API → DB)
4	Availability	Uptime ensured using cloud deployment + failover mechanism (Render/Heroku)	Render, Heroku with auto-restart
5	Performance	Efficient routing, optimized queries, socket-based real-time communication	Caching (optional), CDN (for static assets)

